ORIGINAL RESEARCH ARTICLE

Studies on Power Modulation of Aluminum Smelting Cells Based on a Discretized Mass and Thermal Dynamic Model

CHOON-JIE WONG, JIE BAO, MARIA SKYLLAS-KAZACOS, BARRY WELCH, MOHAMED MAHMOUD, ALEXANDER ARKHIPOV, and NADIA AHLI

With increasing penetration of intermittent renewables, flexible operation of the aluminum smelting process under variable power supply will become critical for smelter operational feasibility and for providing grid demand-side service. As a demand-side power management approach, power modulation dynamically varies the aluminum production rate (thus power consumption) in response to power availability/price. While there were power modulation trials in the industry, many resulted in poor plant performance due to the operational difficulties caused by the spatial variations of process conditions and multi-timescale smelting cell dynamics. This paper presents studies on the feasibility and cell behavior under power modulation based on a first-principles spatially discretized dynamic material and energy balance model. Various combinations of process inputs (line current and interelectrode distance) were used to study the protective side ledge profile, as it should always be maintained for a well-performing, long-lasting cell. Simulation studies of repeated modulation cycles show that ledge does not solidify and dissolve in the same timeframe unless energy is rebalanced, while having slow dynamics thus only stabilizing after a few modulation cycles under a high-frequency modulation. This work contributes to the power modulation modeling and analysis of aluminum smelters, allowing for operational and control optimization.

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I. INTRODUCTION

ALUMINUM plays an important role as decarbonization policies and the shift to a more sustainable world drive a greater adoption of renewable energies and electric vehicles.[1] The reliability of low-cost energy supply is crucial for the aluminum smelting industry, which consumes 8 pct of global industrial electricity,[2] with electricity costs representing more than 30 pct of the total production costs.[3] However, the increased penetration of intermittent renewables also leads to a more variable supply and pricing of electricity.[4] This challenges the smelting industry to compromise between increasing production costs and disrupting the delicate mass and thermal balances of their smelting cells, which impacts cell stability, productivity, and lifespan.[5]

Aluminum smelting cells are traditionally operated at fairly constant power. As a demand-side power management approach, *power modulation* of aluminum smelting cells can dynamically vary the production rate (thus power consumption) in response to power availability/price. With this increased operational flexibility, smelters can be powered by renewable energy and act as virtual ''grid-level storage battery'' by overproducing during periods of low-grid electricity demand to offset lower production during peak demand periods.[6–8] This also allows smelters to offer demand-side response services for the electricity grid, including energy demand-supply balancing and power quality control.[9]

Following increasing interests in demand-side power management of energy-intensive industrial processes,[6,7,10,11] a recent control study[12] investigates the participation of aluminum smelter cells in power grid frequency modulation. While the results are promising, the authors ignored the freezing behavior of the electrolyte and instead opted for simplistic heat loss equations, due to the lack of suitable thermal models. This layer of electrolyte freeze (''ledge'') lining the inner

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cell walls is a crucial aspect of cell design, and plays a very important role in cell performance and longevity by shielding the wall construction materials from the corrosive electrolyte and metal.[13–15] This ledge is sensitive to thermal imbalance and can grow/dissolve, which changes the mass and thermal conditions. The formation of the ledge, and its structure, in response to shifts in thermal balance has been studied in detail in laboratory settings.[15–20] Other researchers focus on controlling the heat loss from the sides of the cell with the use of shell heat exchangers,[21–26] such that operable heat balance and, consequently, an ideal ledge profile can be maintained while power input changes significantly. While such an approach greatly improves the range of power modulation, the cost of retrofitting and difficulty in heat recovery (to maintain a good energy efficiency) could pose additional challenges. Therefore, a spatial dynamic model capable of computing mass and thermal balances, as well as the variations in ledge profile, will be a valuable addition to our repertoire of tools for process monitoring and control.

Successful power modulation of smelting cells requires process engineers to identify safe cell operating range, since most cells were not designed to operate flexibly, especially not with variable line current. Although the industry has experimented with power modulations, many studies have been limited to trial-and-error experiments that come at the expense of unstable cells and degraded cell life,[27–30] due to the operational difficulties caused by the process nonlinearity, spatial variations of process conditions and multi-timescale smelting cell dynamics. Therefore, this warrants a study of the operating range and the dynamics of a cell with the use of a mathematical dynamic model. To the authors' best knowledge, this is the first paper that studies the feasibility and cell behavior under power modulation, including that of a repeated modulation scheme, based on a first-principles spatially-discretized mass and thermal dynamic cell model. While there are simulation studies based on CFD models (*e.g.*, References 31 through 33), those models are too complex to be used for process monitoring and control.

Dynamic models for aluminum smelting cells have been proposed to describe the average cell conditions,[34–45] but the assumptions of cell spatial homogeneity is becoming questionable due to the trends in large modern cell designs, which includes increasing anode size per line current, decreasing electrolyte volume per line current, decreasing interelectrode distance which impedes electrolyte mixing, and increasing alumina feedrate per feeder. With mass and thermal gradients in the electrolyte becoming more significant, and new anodes taking longer to heat up to nominal temperature, lumped parameter models are no longer representative of actual cell conditions. Spatially-distributed models have been proposed,[46–50] but they unrealistically neglect the spatial variations in the mass balance, and by extension, the strong local coupling of mass and thermal balances.

In contrast, the model presented in this article describes the spatially distributed mass and thermal balances, coupled *via* the anode currents and cell voltage. Without requiring prior knowledge of individual anode currents and cell voltage, unlike previous model developments, this model can be used for power modulation studies, including numerical simulations for finding ways to maintain proper material and energy balances for improved cell stability and, as an extension, lifespan. It also enables optimization of cell operations during power modulation based on factors such as power prices and availability.

This paper first presents the existing power modulation studies in the literature, followed by a first-principles spatially distributed dynamic cell model. Steady-state and transient-state studies on the model are then presented, followed by studies of power modulation scenarios.

II. SMELTING CELL FUNDAMENTALS AND EXISTING MODULATION STUDIES

A modern Hall–Hoult smelting cell is illustrated in Figure 1 in which alumina (Al_2O_3), fed by a row of point feeders into a cryolitic melt (Na_3AlF_6 with additives including fluorides and alumina),[14,51] is reduced to aluminum in redox reactions with consumable carbon anodes as follows[52,53]

$$2Al_2O_3(diss) + 3C(s) \longrightarrow 4Al(l) + 3CO_2(g), \qquad [1]$$

$$Al_2O_3(diss) + 3C(s) \longrightarrow 2Al(l) + 3CO(g). \qquad [2]$$

The produced molten aluminum accumulates at the bottom of the cell and replaces the carbon cathode below as the true cathode.[14] The distance from the anode to this metal layer, also called the

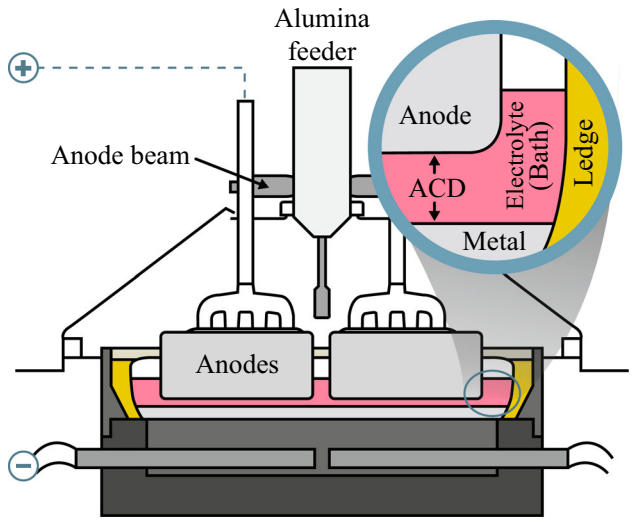


Fig. 1—Schematic of Hall–Héroult cell.

anode-to-cathode distance (ACD), is an important variable in controlling cell heat generation to maintain an operating temperature of near 960 °C. By vertically moving the beam which all the anodes are clamped to, the resistance through this gap can be varied, thus changing the cell voltage and ohmic heat generation.[54] As the cell voltage is controlled within a band to ensure the stability of cell mass and heat balances,[54] and the supplied line current is regulated (constant),[14] this means that the smelting cells are traditionally operated at a fairly constant power. However, when voltage and/or current are being modulated, it becomes challenging to maintain cell balance due to process nonlinearities, strong interactions between mass and thermal balances, and difficulties in measuring and monitoring spatial cell conditions.

The above was the case for Brazillian smelters when the government introduced a variable power tariff that rendered operation infeasible during peak hours due to high energy costs. In 1987 and 1995, Alcanbrasil[27] and Valesul Alumínio,[28] respectively reduced their power inputs without compensating for the energy deficit. As a result, they experienced electrolyte freezing and unstable cell operations; they discontinued power modulation upon obtaining new energy contracts.[29] Since then, other smelters reported to modulate their power supply include Albras in 2001 following Brazil's energy crisis,[55] Alcoa in 2003 following rectifier damages,[56] Hydro in 2002/2003 following reduced water power production,[57] and TRIMET in 2008 due to high power prices.[30] Similarly, many of them reported operational issues and difficulties in process control.

These results are echoed by Biedler's[47] simulation studies on current reduction with a finite-element thermal model. Dupuis[58] noted that this heat deficit can be compensated by raising the ACD (and hence cell voltage); this maintains the heat balance and allows the cell to operate at steady-state, but the power usage is lowered since metal production rate has dropped. In contrast, Stam and Schaafsma[59] took advantage of the cell thermal capacity and implemented a day-night rhythm where the current was lowered at daytime (peak) and raised at night (off-peak). They observed that the electrolyte volume changes slower than its temperature, which showed the challenge in anticipating process dynamics due to its nature of having multiple timescales. Eisma and Patel[60] explained that variations in electrolyte volume are related to the freezing/melting of the "ledge," which is a layer of frozen electrolyte on the walls of a cell (see Figure 1). Based on their experimental results, they concluded that the ledge can build up over repeated modulation cycles, leading to operational issues. This was, however, not corroborated by Reek's[30] experiments, which showed that the ledge profile remained acceptable after 2 months of day-night cycles. In another work,[29] Reek found that the ledge thickness in the metal pad was growing over time, and suggested that the dynamics are time-dependent with an inverse response. Around the same time, Gusberti[31] conducted simulation studies which subject aluminum smelter cells to oscillating power input, and there were no issues in the ledge profile over multiple power modulation cycles. More recently, Ying[61] also performed a series of simulation studies which reduced the current as low as 50 pct of nominal values and for as long as six hours. Although the results indicate successful flexible operation in which the cell can return to its state before the current reduction, the model neglects the freezing behavior of the electrolyte, which would have altered the heat transfer conditions.

These industrial experiments can be costly as poor operational decisions for the complex process affect the entire section of cells *via* the shared line current. Instead, model-based numerical experiments should be explored for low-cost training and operational development purposes.

III. PROCESS MODELING

Cell process variables are highly correlated through the line current and to one another,[31,47,48,62] which leads to the coupling between thermal balance, mass balance, and electricity in the cell.

A. *Thermal Balance*

Thermal balance governs the generation, loss, and transfer of heat at different locations in the cell. A "superheat," which is the difference between the cell operating temperature and the liquidus temperature (the composition-dependent temperature at which the electrolyte starts to solidify), typically of around 10 °C[63] represents the thermal energy available to drive ledge solidification/dissolution, side heat loss, and feed dissolution in a cell.[31,62,64] A high superheat excessively melts the ledge and releases more cryolite (primary phase of the ledge) into the electrolyte,[62,65] changing the electrolyte composition. This further changes the liquidus temperature and, thus, the superheat. This cycle ends when all the parameters in the cell reach an equilibrium,[31,62] where heat generation and heat loss are balanced. The reverse is true for low superheat.

B. *Mass Balance*

Mass balance governs the chemical composition of the electrolyte, taking into account electrolyte flow and feeding of materials. While material feeding is meant to maintain mass balance, experiment shows that it impacts the thermal conditions by removing heat from the local electrolyte region.[66] Additionally, improper feeding rates also affect cell thermal balance through changing electrolyte composition and liquidus temperature.[67,68] The electrolyte composition also changes with the ledge solidification/dissolution, thus affecting thermal balance as explained above.

C. *Electricity*

The electric current and cell voltage both affect and are affected by the local mass and thermal conditions. The regulated line current enters a cell *via* busbars and risers, and is distributed among the parallel network of

anodes. As the voltage drop along each parallel anode-cathode path must be identical, the magnitude of an anode current must vary according to the localized reversible potential, overpotentials, and ohmic voltage drops, all of which arise due to the local conditions of the anode-cathode path (*e.g.*, electrolyte composition and temperature) as per Kirchhoff's Voltage Law.[62,69,70] Kirchhoff's Current Law also needs to be satisfied, with all anode currents summing to line current.[70] Electricity affects the local cell conditions in two ways. Firstly, local material consumption rate depends on the anode current which drives the electrochemical reduction process *via* Faraday's Law of Electrolysis. Secondly, local ohmic heat generation is a function of both anode current and cell voltage.

For smooth operation and high process efficiency, proper cell balances must be maintained.[71] Considering the inevitable disturbances on parts of the cell caused by work practices, as well as the challenges of varying both line current and cell voltage in power modulation, proper cell balances are harder to achieve as there is greater spatial and temporal variability in process variables.[66] Hence, spatially-discretized dynamic models are needed.

D. *Spatially Discretized Thermal Model*

As cell heating occurs primarily *via* the ohmic effect, the distribution of anode currents in a cell readily reflects the distribution of energy input. Therefore, to capture local cell dynamics, the model is discretized according to anode positions and dimensions as demonstrated in Figure 2, similar to our previous work.[62] The electrolyte region is not further vertically discretized in this approach because each current magnitude already represents the total impedance at each anode–cathode path.[62,72] This also eliminates the need to have empirical parameters and relations for electrical resistances in the thermal model. Nevertheless, the lateral heat transfers in the electrolyte and in the metal pad are calculated separately, due to the large difference in their heat transfer coefficients. The anodes are still treated as separate vertical layers, while other vertical layers such as top crust and bottom insulation are treated as thermal resistances and so vertical heat losses can still be computed.

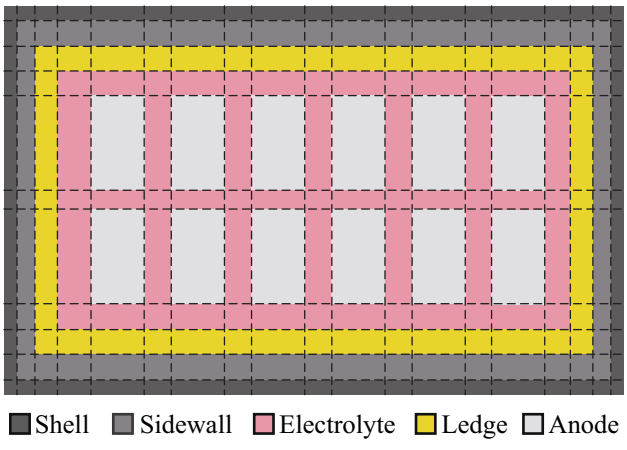


Fig. 2—Discretization of thermal balance model according to anode locations (illustration of a simplified cell with 12 anodes).

The equations governing the thermal transfer in the cell, along with the domains in which they apply, are summarized in Table I. The model parameters specific to the cell technology and design studied in this paper are provided in Table II.

1. *Heat generation*

Virtually all present operating cells are supplied 13 to 16 kWh/kg Al of electricity, which are used as follows:

- Wasteful external heat dissipation, as voltage drops across external cell structures like the busbars and risers.
- Aluminum metal production, using 6.60 kWh/kg Al for a modern technology with a net and gross anode carbon consumption of 0.40 kg/kg Al and 0.50 kg/kg Al respectively. This energy includes the heat for preheating materials at the feeding locations, and thus the energy used for the process at the anodes amounts to 5.474 kWh/kg Al (refer to Table III).
- The remainder is for ohmic heat generation in the cell to maintain thermal balance. This happens mostly underneath an anode as current flows vertically down to the cathode below.[62]

Thus, the local heat generation at each anode-–cathode path, Q_{gen}, is given as

$$Q_{\text{gen}} = I_{\text{an}} \cdot (V_{\text{cell}} - IR_{\text{ext}}) - 5.474\dot{m}_{\text{Al}}(I_{\text{an}}), \qquad [12]$$

where I_{an} is the anode current, V_{cell} is the cell voltage, I is the line current, R_{ext} is the resistance external to the cell that dissipates energy wastefully, and $\dot{m}_{\text{Al}}$ is the hourly rate of metal production given by Faraday's law of electrolysis (Eq. [11]).

2. *Heat transfers and heat losses*

Figure 3 illustrates the heat transfers between various cell domains, as summarized in Table I. The following heat transfers are considered in the model:

1. Heat in the electrolyte and metal are transferred by conduction and advection (induced by the bulk fluid movement). The flow pattern was computed by our industrial partner through computational fluid dynamics studies (e.g., References 76 through 78).
2. Heat is transferred to the ledge surface and anode surface by convection.
3. Heat is conducted up the anodes through their yoke and rod. A portion of this heat is lost by convection to the chamber air enclosed by the cell hoods, while the remaining is conducted further upwards and lost *via* convection to the potroom air.[79]
4. Heat is lost from the crust to the chamber air *via* convection.[62]
5. Heat is transferred down from the metal layer to the cell lining materials (such as vermiculite and firebricks), and then to the shell.
6. Heat is conducted outwards from the ledge to the sidewall, and then to the shell. Heat is also conducted

Table I. Basic Governing Equations in the Thermal Model

Parameter	Domains	Governing Equation	
Thermal Transfer			
Conduction	anode, electrolyte, metal, ledge, sidewall, shell, ledge–sidewall, sidewall–shell, anode–crust, anode–assembly, bottom insulation–shell	$Q = kA\frac{(T_1-T_2)}{L}$ or	(3)
		$Q = \frac{T_1-T_2}{R}$	(4)
		$R = \frac{L}{kA}$	(5)
Convection (Including Radiation)	anode-electrolyte, electrolyte–ledge, metal–ledge, shell–ambient, crust–chamber air, assembly–chamber air, rod–ambient	$Q = hA(T_s - T_\infty)$ or	(6)
		$Q = \frac{T_1-T_2}{R}$	(7)
		$R = \frac{1}{hA}$	(8)
Advection	electrolyte flow	$Q = \rho A v c(T_1 - T_2)$	(9)
Other			
Temperature State Equation	all subsystems	$\frac{d(\rho V c T)}{dt} = Q_{in} - Q_{out} + Q_{gen}$	(10)
Faraday's Law of Electrolysis	anode consumption, alumina consumption, metal production	$\dot{m} = \frac{I_{an} M \eta}{F z}$	(11)

Table II. Key Model Parameters

Material	Parameter	References
Thermal Conductivity, K (W/(m K))		
Anode	4.9 to 6.6	industry
Electrolyte	1000	62
Metal	1000	62
Ledge	1 to 1.5	industry
Graphitized Block (Sidewall)	58 to 77	industry
Silicon Carbide Block (Sidewall)	22 to 31	industry
Steel Shell (Side)	42 to 47	industry
Cast Steel (Yoke)	38.6	industry
Rod	210	industry
Crust	2.6	industry
Collector Bar	30	industry
Firebricks	1.50	62
High-Density Vermiculite	0.30	industry
Low-Density Vermiculite	0.14	62
Steel Shell (Bottom)	31	62
Heat Transfer Coefficient, h (W/(m^2 K))		
Electrolyte—Anode	1100	62
Electrolyte—Ledge	600	industry
Metal—Ledge	1250	industry
Steel Shell (Side)—Ambient	22	industry
Steel Shell (Bottom)—Ambient	20 to 30	industry
Rod—Ambient	40	assumed
Rod/Yoke—Chamber Air	40	assumed
Crust—Chamber Air	20 to 40	industry

between similar subsystems, *i.e.*, ledge–ledge, sidewall–sidewall, and shell–shell.

7. Finally, heat is lost from the outer shell on the bottom and the side to the ambient air *via* convection. Heat loss by radiation is also accounted in the effective heat transfer coefficients for convection.

Additionally, materials are fed into the electrolyte by point feeders installed along the cell center channel (between the two rows of anodes). The energy required to preheat these materials to the operating temperature of near 960 °C and to dissolve them are provided by the local superheat surrounding the feeding zone. This energy requirement per unit mass of feed material can be calculated from the values in Table III.

3. *Ledge dynamics*

The energy needed to change the ledge profile is given by the difference between the heat flux going into the ledge surface from the electrolyte by convection ($Q_{elect\rightarrow boundary}$) and the heat flux going out from the surface to the ledge block by conduction ($Q_{boundary\rightarrow ledge}$):

$$\frac{d(m_{ledge})}{dt} = \frac{Q_{boundary\rightarrow ledge} - Q_{elect\rightarrow boundary}}{\Delta H_{fusion,cryolite}}. \quad [13]$$

Changes to ledge thickness can then be volumetrically calculated using its density. As cryolite is added or removed from the electrolyte as ledge dissolves or solidifies, this changes the mass of the electrolyte, m_{elect}, with

$$\frac{d(m_{elect})}{dt} = -\frac{d(m_{ledge})}{dt}. \quad [14]$$

E. *Spatially Discretized Mass Model*

Our previous works[80,81] have shown that the electrolyte composition of Hall-Héroult cells exhibit spatial variations. This leads to varying heat transfer conditions in different parts of the cell, since the local liquidus temperature (°C) is related to electrolyte composition with[68]:

Table III. Breakdown of Process Energy Required

Energy Use	kWh/kg Al	References
Local Heat Requirement at Feeder Locations	*1.126*	
Preheating Feed (Alumina and Impurities)	0.575	73
Dissolution of Feed	0.551	74
Local Heat Requirement at Anode Locations	*5.474*	
Preheating Anode (Carbon and Sulfur)	0.180	73, 75
Preheating Other Materials and Impurities	0.100	75
Enabling and Completion of Reactions	5.194	difference
Total	6.60	73, 75

Italic value indiates the total values of energy use in each category.

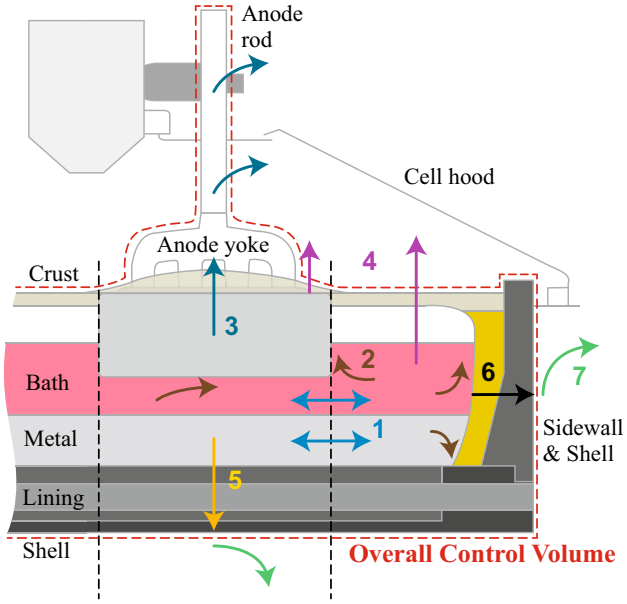


Fig. 3—Illustration of some important heat transfers considered in the model.

$$T_{liq} = 1011 + 0.5C_{AlF_3} - 0.13C_{AlF_3}^{2.2} - \frac{3.45C_{CaF_2}}{1 + 0.0173C_{CaF_2}} + 0.124C_{CaF_2}C_{AlF_3} - 0.00542(C_{CaF_2}C_{AlF_3})^{1.5} - \frac{7.93C_{Al_2O_3}}{1 + 0.0936C_{Al_2O_3}} - 0.0017C_{Al_2O_3}^2 - 0.0023C_{AlF_3}C_{Al_2O_3} - \frac{8.9C_{LiF}}{1 + 0.0047C_{LiF} + 0.001C_{AlF_3}^2} - 3.95(C_{MgF_2} + C_{KF}), \quad [15]$$

where all the C concentration terms in this equation are in wt pct. In a typical Al_2O_3–AlF_3–CaF_2–Na_3AlF_6 system, the liquidus temperature can be accurately determined by just considering the dynamics of Al_2O_3 and AlF_3 concentrations, since CaF_2 concentration typically remains relatively stable or slow-varying compared to other species, and it has the least impact on liquidus temperature. This is demonstrated by Figure 4, which shows the liquidus temperature as a function of alumina, aluminum fluoride, and calcium fluoride concentrations (calculated using Eq. [15]) in the ranges typically observed in an industrial cell.

Table 4. Important Initial Values Used

Parameter	Initial Value
$C_{Al_2O_3}$	3.0 wt pct
C_{AlF_3}	11.9 wt pct
C_{CaF_2}	6.4 wt pct
C_{MgF_2}	0 wt pct
C_{LiF}	0 wt pct
C_{KF}	0 wt pct
T_{elect}	968 °C
T_{ledge}	820 °C
$T_{sidewall}$	600 °C
L_{ledge}	5 cm

In a cell, there are feeders that each deliver an isovolumetric shot of material when activated. The shots are assumed to have distributed among the electrolyte regions surrounding the feeding zone. This is because the solid materials can float and disperse on the electrolyte surface as a 'raft' for short periods.[82]

1. *Alumina concentration*

The dynamics of alumina dissolution can be expressed with two first-order mechanisms.[47] The first models the largest portion of alumina power, r (0.75), that dissolves quickly in the bath with rate constant k_f (0.099 s^{-1}), while the second models the remainder of alumina that forms agglomerates which dissolve slowly at a rate constant k_s (0.005 s^{-1}).[47] Therefore, for each subsystem, the concentrations of solid alumina that dissolves quickly, C_f, and slowly, C_s, are given by

$$\frac{dC_f}{dt} = -k_f C_f + \frac{r\dot{m}_{fed}}{m_{elect}}, \quad [16]$$

$$\frac{dC_s}{dt} = -k_s C_s + \frac{(1-r)\dot{m}_{fed}}{m_{elect}}, \quad [17]$$

where $\dot{m}_{fed}$ is the mass feeding rate of alumina in the subsystem and all the concentration terms are expressed as wt/wt. The dissolved alumina dynamics are composed of: 1. Feed dissolution (Eqs. [16] through [17]).

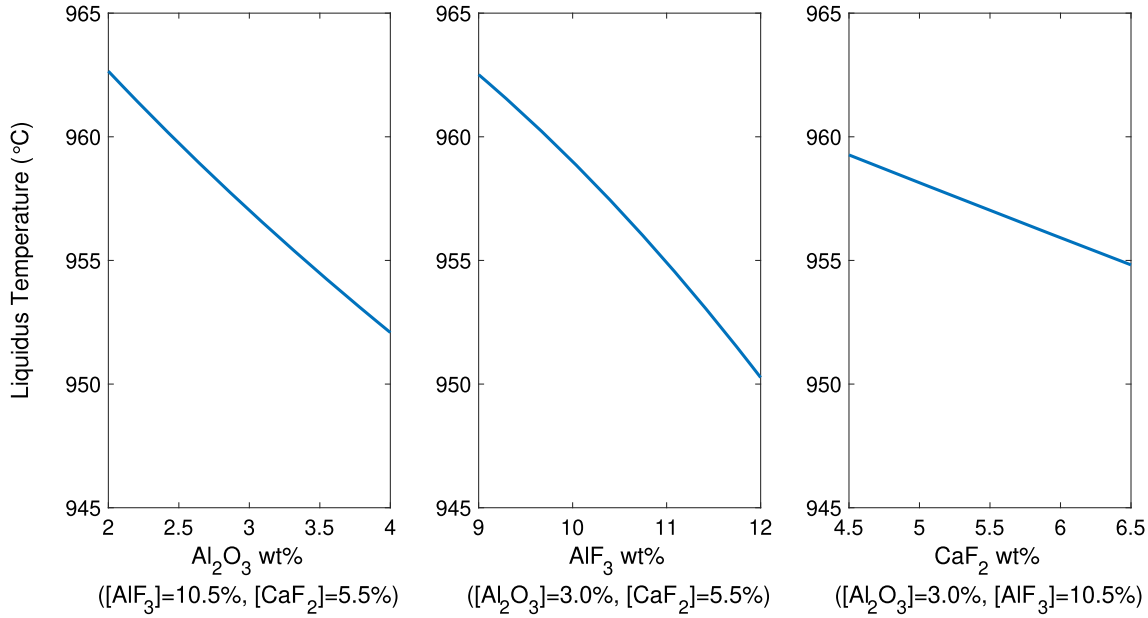


Fig. 4—Liquidus temperature as a function of alumina, aluminum fluoride, and calcium fluoride concentrations in ranges typically observed in an industrial cell.

2. Consumption in the process reactions (Eqs. [1] and [2]) with a rate of $\dot{m}_{\text{consum}}$ modeled with Faraday's law of electrolysis (Eq. [11]).
3. Advective mass transfer due to flow of bulk electrolyte (Eq. [9]) with a rate of $\dot{m}_{\text{flow}}$.
4. Change in electrolyte composition due to the solidification or dissolution of ledge (the composition of which is primarily cryolite). Solidification of ledge decreases electrolyte mass and increases alumina concentration, and *vice versa*.

Therefore, the dissolved alumina in each subsystem can be computed with

$$\frac{\mathrm{d}(C_{\mathrm{Al_2O_3}})}{\mathrm{d}t} = k_{\mathrm{f}}C_{\mathrm{f}} + k_{\mathrm{s}}C_{\mathrm{s}} - \frac{\dot{m}_{\text{consum}}(I_{\text{an}})}{m_{\text{elect}}} + \frac{\dot{m}_{\text{flow}}}{m_{\text{elect}}} - \frac{C_{\mathrm{Al_2O_3}}\frac{\mathrm{d}(m_{\text{elect}})}{\mathrm{d}t}}{m_{\text{elect}}}. \quad [18]$$

2. *Fluoride concentrations*

Aluminum fluoride is only fed to alter liquidus temperature of the electrolyte and to make up for fluoride losses from the cell. Therefore, the cell studied in this paper only requires about 50 kg daily, in contrast to about 6 tonnes of alumina daily. As the aluminum fluoride concentration remains fairly constant due to periodic feeding, the model is simplified by assuming that the fed aluminum fluoride instantly distributes and dissolves in the electrolyte region surrounding the feeding zone. The mass balance components are then:

1. Mass of aluminum fluoride introduced to the electrolyte, $\dot{m}_{\text{fed,lF}_3}$.
2. Advective mass transfer due to flow of bulk electrolyte (Eq. [9]) with a rate of $\dot{m}_{\text{flow,AlF}_3}$.
3. Change in electrolyte composition due to the solidification or dissolution of ledge.
4. Fluoride emissions due to hydrolysis of electrolyte and vapor $\dot{m}_{\text{G,F}}$, electrolyte volatilization $\dot{m}_{\text{V,F}}$, and electrolyte entrainment $\dot{m}_{\text{E,F}}$. Complete details of these relations have been published elsewhere.[44,83,84]

Hence, the dissolved aluminum fluoride in each subsystem is given by

$$\frac{\mathrm{d}(C_{\mathrm{AlF_3}})}{\mathrm{d}t} = \frac{\dot{m}_{\text{fed,AlF}_3}}{m_{\text{elect}}} + \frac{\dot{m}_{\text{flow,AlF}_3}}{m_{\text{elect}}} - \frac{C_{\mathrm{AlF_3}}\frac{\mathrm{d}(m_{\text{elect}})}{\mathrm{d}t}}{m_{\text{elect}}} - \frac{\frac{\mathrm{Mr(AlF_3)}}{3\mathrm{Mr(F)}}\left(\dot{m}_{\text{G,F}} + \dot{m}_{\text{V,F}} + \frac{C_{\mathrm{AlF_3}}}{1-C_{\mathrm{Al_2O_3}}}\dot{m}_{\text{E,F}}\right)}{m_{\text{elect}}}, \quad [19]$$

where $\mathrm{Mr}(\cdot)$ is the molar mass or atomic mass.

F. *Anode Current and Cell Voltage*

As a result of Kirchhoff's Laws, the voltage drop across each anode-cathode path must be identical, while all anode currents must sum to the line current. Therefore, the cell voltage and N anode currents can be simultaneously computed from

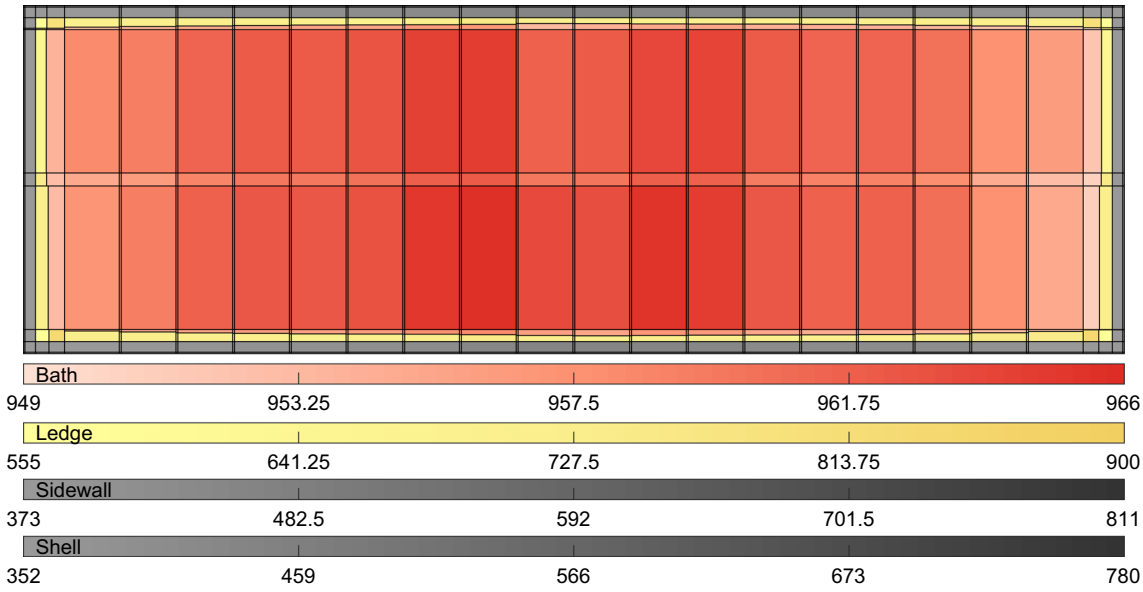


Fig. 5—Heat map of thermal balance variables in equilibrium (°C).

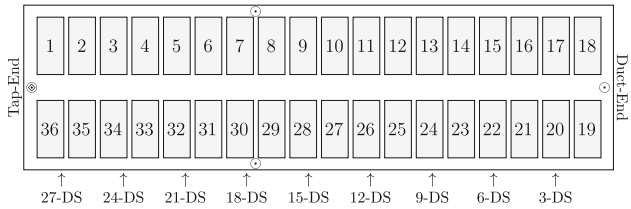


Fig. 6—Measurement locations for model validation (⊙ = electrolyte temperature, ↑ = shell temperature, ⋄ = electrolyte sample).

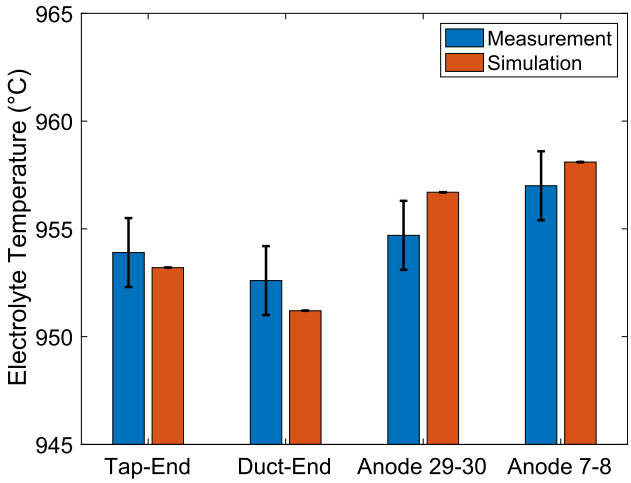


Fig. 7—Comparison of measured and simulated electrolyte temperature. Error bars represent measurement uncertainties.

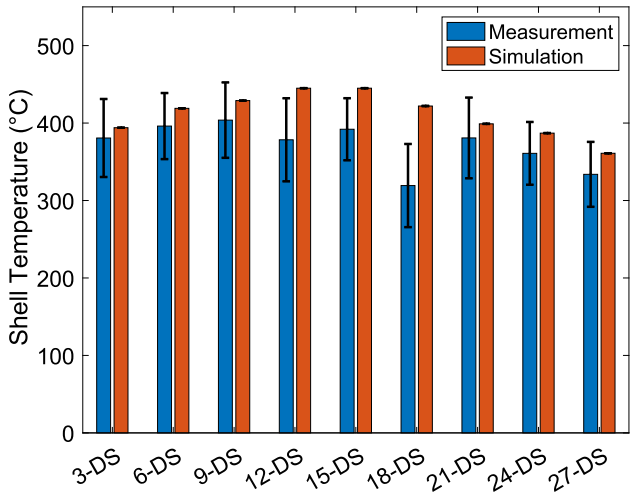


Fig. 8—Comparison of measured and simulated shell temperature. Error bars represent standard deviation.

$$\begin{cases} V_{\text{cell},1}(I_{\text{an},1}) & = V_{\text{cell},2}(I_{\text{an},2}) \\ V_{\text{cell},1}(I_{\text{an},1}) & = V_{\text{cell},3}(I_{\text{an},3}) \\ & \vdots \\ V_{\text{cell},1}(I_{\text{an},1}) & = V_{\text{cell,N}}(I_{\text{an,N}}) \\ I & = \sum_{n=1}^{N} I_{\text{an,n}} \end{cases}, \qquad [20]$$

where the cell voltage empirical relations, V_{cell}, are well-established and have been published elsewhere in detail.[70,85] In short, the cell voltage consists of the

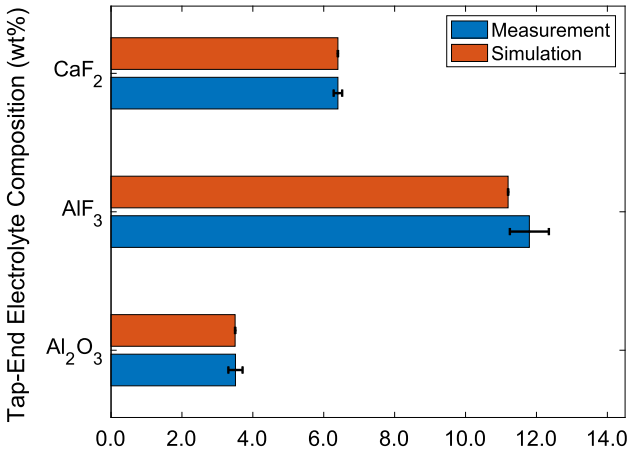


Fig. 9—Comparison of measured and simulated electrolyte composition at the tap-end. Error bars represent standard error.

reversible potential, overpotentials due to concentration gradients and surface reactions at the electrodes, as well as ohmic potential drops through the electrolyte, bubble, anodes, cathodes, and external superstructures.

G. *Simulation and Model Validation*

Due to the strong coupling between process variables, Eqs. 4 through 20 must be solved simultaneously, using initial values in Table 4 for each model subsystem. The model was simulated for 50 hours to let all the variables reach stable values, which are shown in Figure 5. To validate the discretized model, measurements were taken from various locations on an industrial cell as shown in Figure 6, including electrolyte temperatures (HERAEUS FIBERLAB, bath temperature mode), steel shell temperatures (FLUKE Ti32 infrared thermometer, emissivity 1.0), and electrolyte composition (PANALYTICAL CUBIX X-ray diffraction, Rietveld software).

Figures 7 through 9 compare the cell measurements to the open-loop simulation results. Figure 7 shows that the electrolyte temperatures predicted by the discretized model reflect well the measurements from various locations in an industrial cell. The differences observed are probably due to uncertainties in initial ACD values, which affects the path resistance, and therefore, the anode current and local heat generation. While the model assumes uniform ACDs as initial values, the actual ACDs of anodes 29 and 30, for example, could be higher than average, which decrease the anode currents and lower the heat generation in the areas, resulting in lower bath temperatures than expected. This can also be seen in Figure 8, where the location 18-DS has abnormally low shell temperature. This also causes neighboring shell temperatures to deviate from expected values. Additionally, the discrepancies in shell temperatures are also caused by their slow dynamics, as heat from within the cell must pass through multiple insulation layers before reaching the shell. With frequent cell operations, the measured shell temperature will not align with the quasi-steady state values (*e.g.*, bath temperatures) inside the cell. Regardless, this model is still useful as uncertainties in model parameters and initial conditions can be rejected by using such a model in an online, closed-loop estimation algorithm that continuously measures and minimizes the estimation error, provided that the system is observable for the measurements, as explained in a previous work.[86] Lastly, Figure 9 shows a good agreement between measured and simulated electrolyte composition at the tap-end region. The samples were only taken from one location as it is logistically challenging to capture all the data in the same time window. However, our previous works[70,87] have confirmed that the mass balance equations[47] can accurately compute the changes in alumina concentration.

Although the discretized cell model was only validated under nominal "steady-state" conditions, it can still allow for further behavioral studies of local cell conditions, as it was derived from the first principles. As these local cell conditions are not conventionally measured or monitored, the model becomes a valuable tool for conducting and studying power modulation schemes.

IV. STEADY-STATE MODEL RESPONSE STUDIES

As explained in Section III–D–1, only a portion of the supplied electrical power is used to generate heat. This heating power should be kept constant if the cell thermal balance is to be maintained under different sets of controlled variables (ACD and line current in a typical modern cell). Although aluminum fluoride concentration is another possible variable to be manipulated for adjusting cell heat balance, it is not in the scope of these studies for two reasons:

1. Its induced thermal response is indirect and very slow, since it has to first change the liquidus temperature and superheat.
2. Most smelting cells are built with aluminum fluoride feeders but not sodium oxide and calcium oxide feeders.[88,89] This means that fluoride concentration cannot be decreased by neutralization with the oxides *via* automatic control.

To illustrate how line current and ACD affect the cell voltage and power usage, the steady-state model response was first qualitatively studied by considering only the cell voltage equation and the total production energy requirement of 6.60 kWh/kg Al, while assuming most other variables remaining constant. This study excludes the impacts of the spatial distribution nature and the coupling between cell states.

Figures 10(a) and (b) show the effects of varying ACD. As this effectively changes only the resistance of the electrolyte layer, it can be seen in Figure 10(a) that the ohmic voltage component changes linearly with ACD, while other voltage components remain constant. This includes the reversible potential, as current efficiency is assumed to remain constant. As a result, the

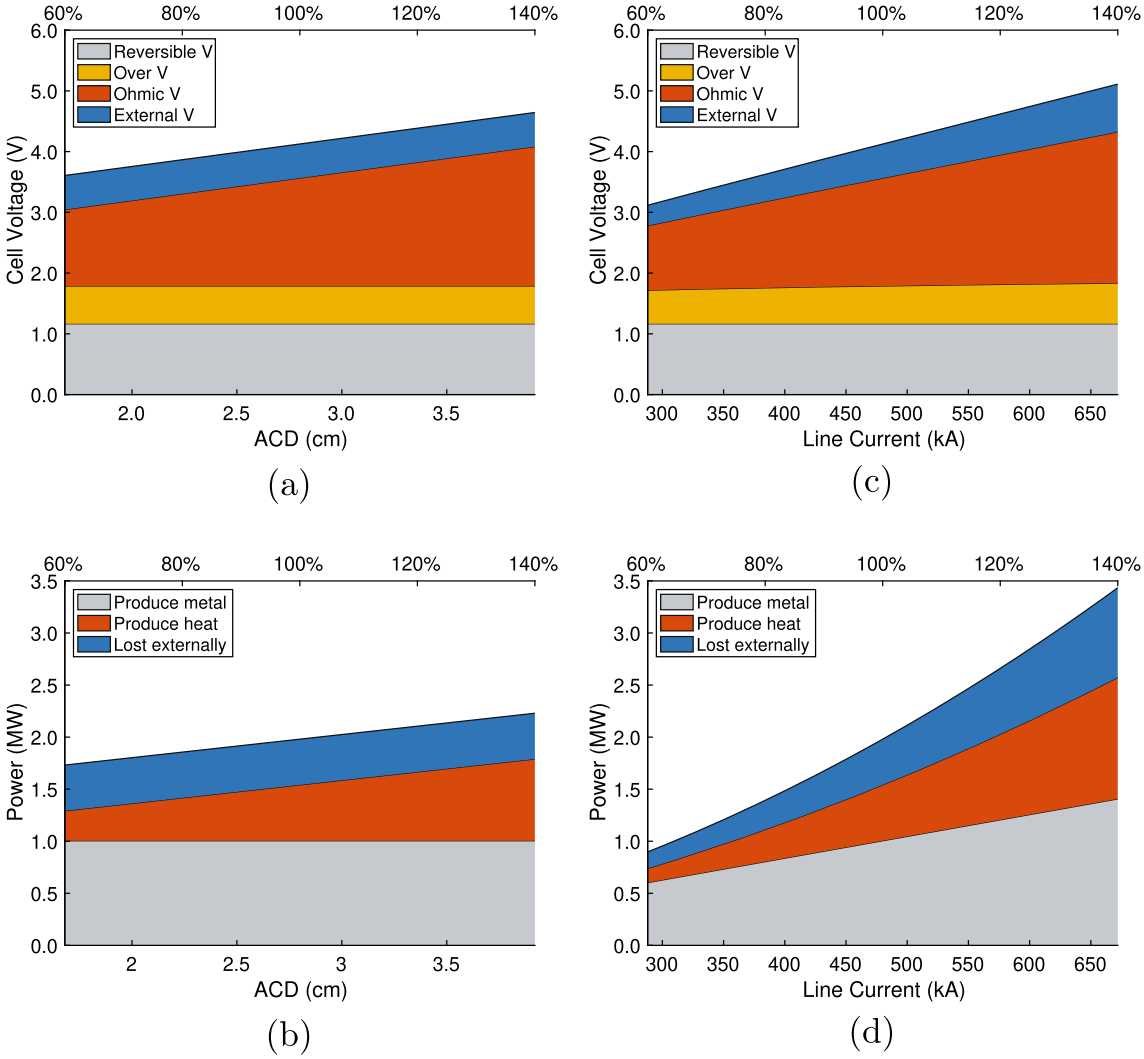


Fig. 10—Impact of varying (*a*) ACD on cell voltage components, (*b*) ACD on power usage components, (*c*) line current on cell voltage components, (*d*) line current on power usage components.

power used to produce heat also increases linearly with ACD, as shown in Figure 10(b). This is expected as raising anode beam is one of the conventional procedures to raise electrolyte temperature.

Figures 10(c) and (d) show the effects of varying line current. Line current affects all the voltage components. From Ohm's Law, the ohmic and external voltage drop components increase with current. The overpotentials are also functions of current, but they increase only slightly and are not visible in Figure 10(c). Figure 10(d) shows that increasing line current greatly raises the power consumption, because metal production rate increases linearly to line current while heat generation increases quadratically ($P = I^2R$). Figure 10 also shows that for the same variation (± 40 pct) in ACD and line current, power is much more effectively modulated with the latter. However, more power is wasted through external voltage drops at a higher line current, while there is a diminishing power reduction at lower current. On the other hand, raising ACD increases power consumption for cell heating without increasing metal production.

When either ACD or line current is varied, the cell thermal balance is disrupted. Inspired by Dupuis' work,[58] Figure 11 shows the ACD required for each line current value such that the heat generation and cell thermal balance are maintained, while the total power consumption is still varied. Unfortunately, this strategy is limited for two reasons. Firstly, by always maintaining heat balance, the cell thermal capacity is not utilized, limiting the range of modulation. Second, this strategy is more suitable for finding a new steady-state operating regime rather than for automatic control, due to the quadratic changes in ACD. As modern cells have high anodes-to-electrolyte volume ratios, electrolyte height is sensitive to ACD changes. Thus, large increases in ACD can pull the anodes out of the electrolyte, which has to be remedied by adding electrolyte into the cell. Alter-

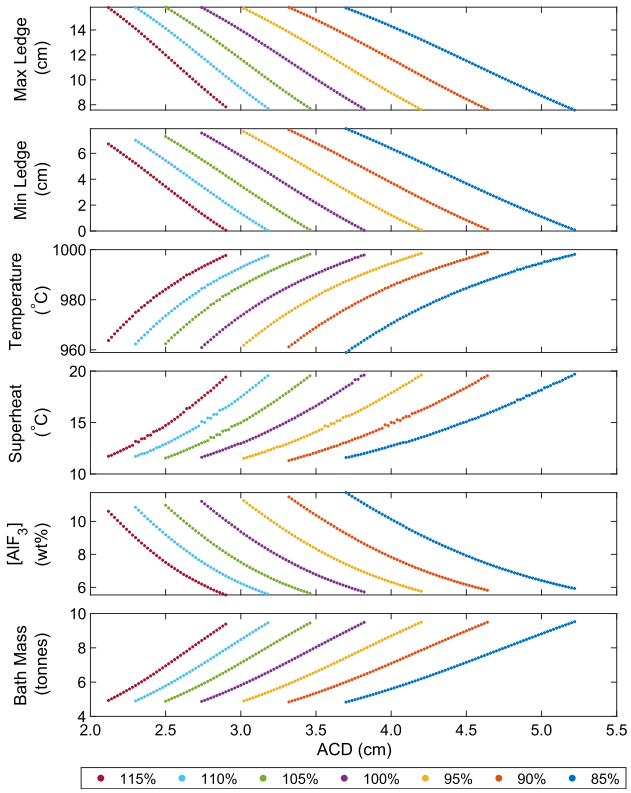


Fig. 12—Operating range for an industrial cell under different combinations of ACD and line current (expressed in pct of nominal line current).

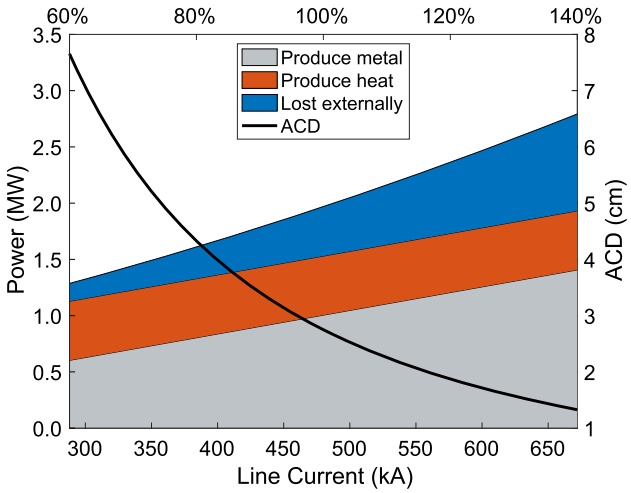


Fig. 11—Cell power usage by varying both ACD and line current while maintaining cell heat balance.

natively, the thickness of anode cover insulation can be increased for greater heat retention. Both operations are manual, limiting how frequently and how many cells can be worked on. Additionally, low ACD can lower process current efficiency due to anode-cathode short-circuiting, as well as the increase in back reaction due to proximity of cathode metal and anode gases.[29]

V. TRANSIENT MODEL RESPONSE STUDIES

The cell thermal capacity arises from the cell construction materials, the electrolytic bath materials, as well as the ledge, which helps to maintain heat balance by absorbing or releasing latent heat to the electrolyte. Therefore, while ensuring that the ledge profile stays within physical bounds, the heating power can be allowed to fluctuate as power is modulated. These operational limits can be determined by simulating the cell model [Eqs.(4)–(20)] under various combinations of line current and ACD.

The final steady value of some important variables under different line currents and ACDs are shown in Figure 12. For a given line current, there is an operationally acceptable range of ACD. Due to the nonlinearity in heat generation, ACD has a wider range and gentler line gradient at lower line current, to have the same change in heating power. Conversely, variables are more sensitive to ACD changes at higher currents, which can make control more challenging. For small changes in line current, Figure 12 is also a useful reference to find sets of ACD and line current which can maintain the thermal variables. For example, a horizontal line can be drawn for any of the variables, and the ACD values can be read from the intersection points. Two such pairs are [100 pct current, 2.75 cm ACD] and [95 pct current, 3.0 cm ACD].

A dynamic simulation with day-night rhythm, where the ACD and line current were switched between these two sets of values every 12 hours, confirmed that cell variables remain constant throughout the modulation period due to constant heating power, as shown in Figure 13(a). Since it is very important for the ledge profile to stay within bounds, its maximum, minimum, and average values are plotted in the figure; average values were plotted for other variables for easier reference although the distributed states and parameters were accounted for in the model. The oscillating alumina concentration and cell voltage were due to conventional feeding algorithm,[90] and the slight difference in the alumina concentration level, as power was modulated, was caused by feeding rate adjustments according to the line current. The modulation amplitude was limited as the heat deficit introduced by lowering current must be compensated by raising ACD.

A contrasting strategy is to fully utilize the latent effect of the ledge and other heat capacities of the cell, without ACD adjustments. For example, Figure 13(b) shows the modulation of current between 100 and 115 pct (ACD unchanged at 2.75 cm). This study uses a larger magnitude of current modulation to highlight the role of the ledge in minimizing the thermal imbalance through latent heat absorption/release. Without balancing the heating power, the cell mass and thermal variables were observed to fluctuate throughout the modulation cycles. Figure 13(b) also shows the different delays in thermal responses of the electrolytic bath and the ledge. The ledge thickness reached steady-state after about 12 hours, which matches Gusberti's studies,[31] while the electrolyte temperature quickly approached its final steady value only after about 5 hours, partially due

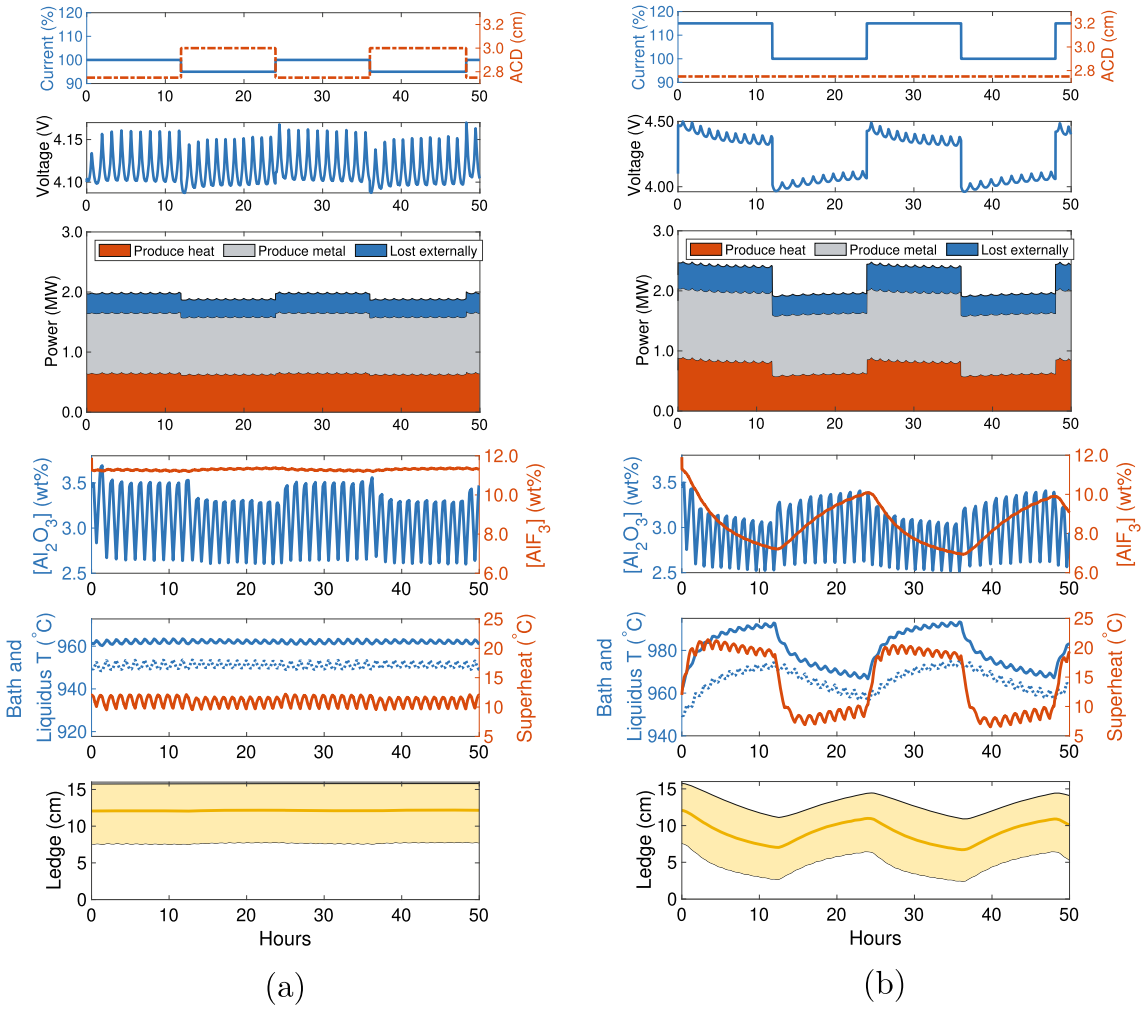


Fig. 13—Behavior of cell states and variables under power modulation with (*a*) ACD/voltage compensation for constant heat generation, (*b*) utilization of cell thermal capacity.

to the latent heat regulation from the ledge. However, the electrolyte temperature continued to evolve slowly after that as a response to the slow variation in liquidus temperature caused by the slowly evolving ledge profile.

The simulations shown in Figure 13 represent two extreme strategies in terms of cell thermal balance, but the model allows studies of any combination of line current and ACD as process inputs. With the model, it is also possible to compute the optimal control actions to maximize profit based on power availability, power costs, utility green tariffs, cost of raw materials, aluminum selling price, energy storage (battery) costs and cycle life, and so on. This is outside the scope of the present work, but an example simulation study is provided in Figure 14 to illustrate one possible operational strategy. In this study, the hourly Australian photovoltaic power output (Reference 91, MERRA-2 dataset, averaged 2019 data) and ambient temperature profile (Reference 92, 2–3 Jan 2022, summer) were considered. The line current was increased during day-time as solar power becomes available, with the target line current, I, computed implicitly from

$$P_{\text{baseload}} + P_{\text{solar}}(t) = I(t) \cdot V_{\text{cell}}(I(t), C(t), T_{\text{elect}}(t)), \quad [21]$$

where P_{baseload} is the nominal cell power consumption and P_{solar} is the photovoltaic power output. The excess accumulated heat was released at night, when the ambient temperature is lowest and the rate of heat release from the cell is highest. Therefore, the period of low current (and low metal production) was shortened to maximize production.

In this study, it was assumed that the smelter has a fixed-price electricity contract, with the same energy cost during off-peak hours. The ACD was also not manipulated to keep the study simple. Other strategies are possible depending on cell design and smelter plant infrastructure. For example, if a plant generates power primarily with gas turbines, it would be better to increase line current at night when ambient temperature is low, since gas turbines will have higher efficiency.[93] Additionally, resistance of external cell components such as busbars also decreases with ambient temperature, and thus the portion of power wasted externally

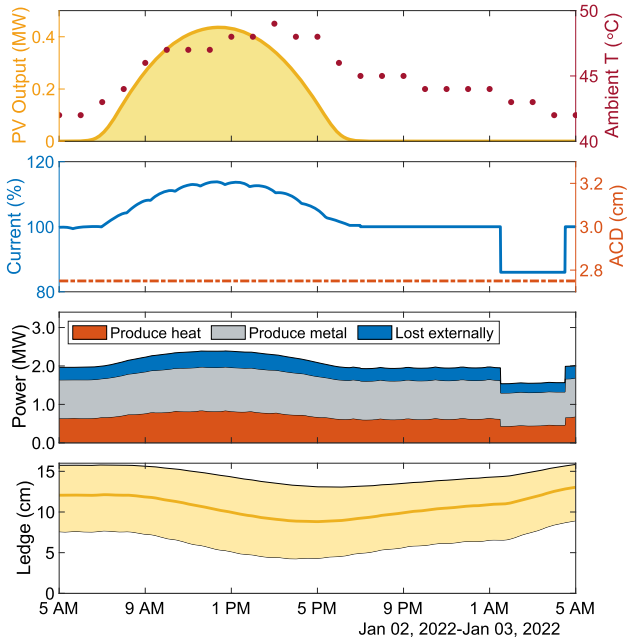


Fig. 14—Behavior of cell states and variables with additional solar power supply during the day, and shorter period of heat release at night when ambient temperature is low.

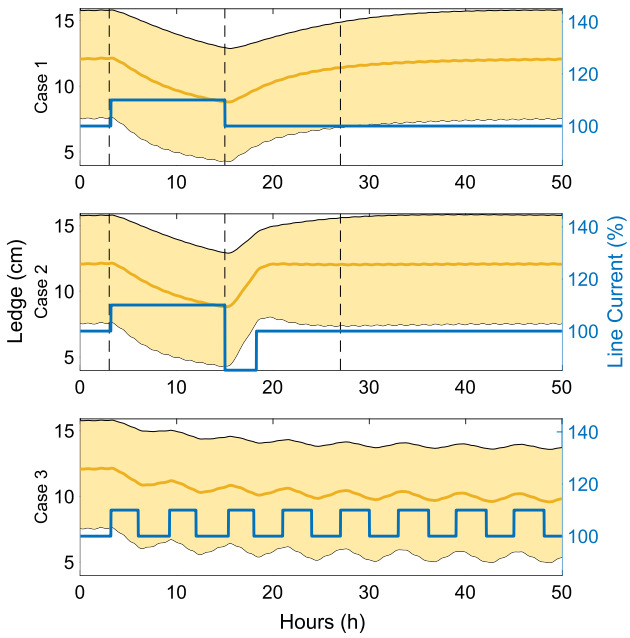


Fig. 15—Study of ledge dynamics under various line current profiles. Case 1: Power increased for 12 h. Case 2: Power increased for 12 h and decreased for 3 h for energy balance. Case 3: Power modulated between high and nominal levels, each for 3 h.

would reduce, and more power can be used for metal production and for heat balance. Increasing power consumption late at night also acts as a good industrial demand-side response which helps improve power grid stability. With the proposed model, more numerical experiments and studies like these are possible.

VI. REPEATED MODULATION CYCLES

The stability of ledge profile over repeated modulation cycles is very important for proper cell operation. However, although inconclusive, it has been suggested in the literature that the ledge can grow unboundedly or deplete completely over cycles. Due to the strong coupling effects between variables and nonlinearity in the system, it is possible that the previous equilibrium point cannot be returned to after each modulation cycle. For example, the melting of ledge may irreversibly affect the liquidus temperature and superheat of the cell. Therefore, three case studies were performed to study ledge reversibility, and the results are presented in Figure 15.

Case 1 temporarily increases the line current by 10 pct for 12 hours, comparable to Eisma and Patel's experiment.[60] Similar to their findings, the ledge dissolved quickly as the power was raised, but upon restoration of the power level, the ledge grew slowly and did not return to its original thickness in the same timeframe of 12 hours. However, this does not mean the ledge will deplete through modulation cycles; it eventually reached the original thickness about 30 hours after power level restoration. This difference in cell response is due to the nonlinear nature of the system; with the momentary introduction of surplus energy, there is a net energy intake, and therefore, ledge dissolution happens quicker than re-solidification. The reverse is also true: if the power is first decreased and then restored, there is a net energy deficit in the system, and thus the rate of ledge solidification is quicker than the re-dissolution.

Case 2 shows that by increasing and decreasing the power input, such that there is zero net change in energy, the ledge profile can be restored much more quickly. To emphasize the distributed nature of the process variables, the power was decreased for only 3 hours (85 pct line current) for net energy balancing. In this timeframe, the average ledge thickness has reached its original value. However, it is seen that thicker ledge, having higher thermal mass, responded slower than the thinner areas which exhibited an inverse response. This highlights the importance of having a discretized model to capture local cell dynamics.

Finally, Case 3 shows the effect of short power modulation periods relative to the response time of the ledge. While the ledge needs about 12 hours on average to reach its new equilibrium value, a complete modulation was cycled every 6 hours. With the power being modulated between high and nominal levels, there was an imbalance in net energy over time. The ledge became thinner but eventually stayed within some bounds. Conversely, if power modulation is cycled between low and nominal levels, the ledge will grow and may stay within bounds, depending on the magnitude and period of low levels. This may not be immediately obvious if the ledge was only experimentally measured for the first few cycles, from which one would conclude the ledge will keep growing or depleting. While this case study investigates the effect of net energy imbalance, it is also relevant for experimental studies that introduce high and low power swings with the hope of balancing the net

energy input. In reality, it is difficult to balance the net energy with control actions calculated offline, due to the lack of process observability as well as uncertainties in parameters of this nonlinear system. Therefore, it is important that a closed-loop feedback approach be considered when modulating the power, perhaps utilizing 'soft sensors' based on state estimation, such as a Kalman Filter.[86]

In closing, it is also important to note that the melt-freeze is a complex phase system and its behavior can differ under transient state. While previous experimental results[15] have shown that a dense sealing primary phase cryolitic layer forms at the electrolyte-ledge interface, which indicates that the interface temperature is at liquidus temperature, this layer is not available during rapid freezing or melting. Instead, porous dendritic structures are exposed to the electrolyte which can alter local freezing or melting behavior.[94,95] Additionally, ledge samples from industrial cells and laboratory ''cold-finger'' experiments also show small amounts of other components, such as CaF_2 and carbon dusts.[15,19,20] All these can lead to discrepancies in the modelled dynamic response under power modulation schemes, and thus should be accounted for with appropriate equivalent heat transfer coefficients.

VII. CONCLUSION

Proper cell balances are essential for low process faults and high process efficiency, but they can be severely disrupted by power modulation efforts, which are either motivated financially or necessitated by power shortages. This paper presents a spatially-discretized first-principles mass and thermal model of an aluminum reduction cell, complete with cell voltage and individual anode currents. It applies to a wide selection of prebaked smelting cells, as cell technology and design are accounted for by the model parameters. As this generalized model describes conditions under normal operations, as well as power modulation, without requiring prior knowledge of anode currents and cell voltage, model-based studies can replace trial-and-error experimental studies, as well as allow future work on optimization and control of the line current and ACD, which will reduce training costs as well as risk of process faults and cell damage.

Both steady-state and transient-state analyses were conducted with the model. Steady-state analysis highlights the difference between manipulating the ACD and the line current. Line current has a larger impact on cell power consumption, since it drives the electrochemical reaction and is responsible for ohmic heating of the entire electrical path. On the other hand, transient-state analysis more completely shows the interactions between the different states as well as between the same state distributed spatially, including the following:

- At higher line current, the range of ACD which allows for the ledge thickness to stay within bounds is smaller. This makes control at higher line current more challenging, so a model-based approach to control will be beneficial.
- When line current is modulated, ACD can be changed to compensate for changes in heating power. However, values of ACD are bounded by operational limits. Instead, the heat capacity of the cell can be utilized to allow the thermal variables to swing.
- Studies of repeated modulation show that, with a temporary change in power supplied, the ledge does not restore to its original thickness, confirming past experimental studies.[60] This is because the net energy input is not balanced. However, differing from conclusions in the literature, the ledge does restore slowly over time.
- Due to the lag in ledge response, a repeated modulation at a relatively high frequency can cause the ledge thickness to only approach its final value after several modulation cycles.

Future improvements will focus on model-based optimization to derive the most profitable ACD and line current trajectory inputs based on energy storage options, as well as electricity price and availability. Due to the serial connection in smelting cells, the strategies to power modulation would need to involve the entire cell section. One such strategy is to vary the line current as the primary goal, whilst ensuring that the ACD of individual cells still stay within feasible operating range. However, the proposed discretized model is also ready for future cell designs with individual ACD control or with shell heat exchangers,[21,23] which would allow for even more extreme power modulation strategies.

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CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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